Tribhuvan University

Faculty of Humanities & Social Sciences

OFFICE OF THE DEAN

Digital Logic

BCA — Semester I — 2022 — Answer Sheet

Subject Code: CACS103

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

2. Subtract: 11101.11110100.101 using both 1's and 2's complement.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Simplify given Boolean function in both SOP and POS using k-map where d represents don't care condition.F (A, B, C, D) = ∏(0, 1, 3, 7, 8, 12) and ∏ d(5, 10, 13, 14)

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. Define combinational circuit. Write a combinational circuit design procedure. Design a half adder with its truth table, logic diagram and Boolean expression.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Define Priority encoder. Design 4:2 priority encoder with its block diagram, truth table, circuit diagram and mathematical expression.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Why NAND and NOR gates are called universal gates? Realize NAND gate as universal gate.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. What is race around condition? Explain how JK master flip flop is used to eliminate race around condition.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Differentiate between synchronous and ripple counter. Design mod 7 ripple counter with its state diagram, sequence table, logic diagram and timing diagram.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Group C

Attempt any TWO questions[2x10=20]

9. Define PLA with its block diagram. Realize BCD to gray code converter using PAL.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. What is register? Explain types of registers depending on input output with its block diagram & logic diagram.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. What is state diagram? A sequential circuit with two D flip-flops A and B, one input x and one output z is specified by the following next state and output equations:A(t + 1) = A' + B, B(t + 1) = B' x, z = A + B'i) Draw the logic diagram of the circuitii) Draw the state tableiii) Draw the state diagram

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.